Equivalent Ratios

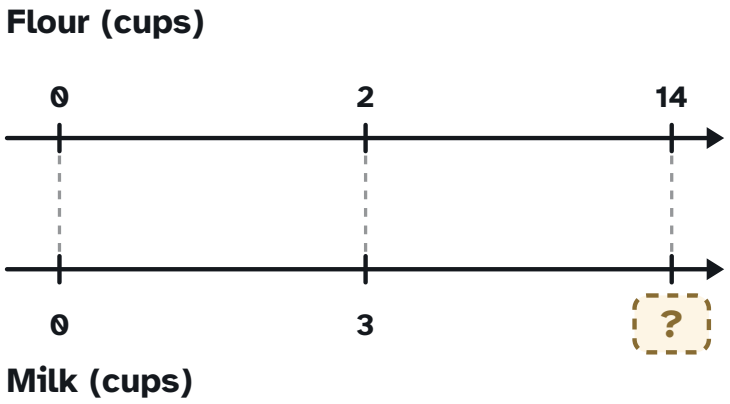
Find the scale factor that grows a recipe and keeps the ratio the same.

Name _____ **Date** _____

Tonight's spark

A lemonade recipe uses 3 cups of water for every 2 cups of juice. A team fills a cooler with 18 cups of water. How many cups of juice do they need?

Make a guess first: more than 10 cups of juice, or fewer?



Our class model: the scale factor

Try it

Work it out on paper. Write one answer on each line.

1. **A recipe uses 3 eggs. You want to use 12 eggs. What is the scale factor?**

   A. 9 **B.** 4 **C.** 15 **D.** 36

   My answer _____

2. **Rice to water is 1:2. You use 6 cups of rice. How many cups of water?**

   A. 12 **B.** 8 **C.** 3 **D.** 6

   My answer _____

3. **A smoothie uses 2 bananas and 5 strawberries. With 8 bananas, how many strawberries?**

   A. 13 **B.** 10 **C.** 20 **D.** 40

   My answer _____

4 **Which action keeps a recipe tasting the same when you make 3 times as much?**

A. Add 3 to each amount **B.** Multiply only one amount by 3

C. Subtract 3 from each amount **D.** Multiply every amount by 3

My answer _____

5 **Lemonade is 3 lemons for 6 cups of water. How many lemons for 24 cups of water?**

A. 18 **B.** 21 **C.** 12 **D.** 4

My answer _____

6 **Divide both numbers of 18:24 by 6. What ratio do you get?**

A. 12:18 **B.** 3:4 **C.** 4:3 **D.** 3:24

My answer _____

Talk about it

What is the scale factor, and how did you use it on the second ingredient?

Home mission

Use a pitcher and a cup. Pretend a drink needs 3 cups of water for 2 cups of juice. Ask your student: how much juice for 9 cups of water? What number did you multiply by?

If we get stuck, ask

- What number do we multiply the first term by to reach the new amount?
- Did the other term get multiplied by that same number?
- Does the new ratio simplify back to the old one?

Do it on a screen

Scan or type the link. Same lesson, with hints.

eduwonderlab.com/lessons/3-9-part2/homework.html

How did it feel? (optional)

Tricky Getting there Got it